Initial Progress & Proposal Report

SENG 495 Senior Graduation Projects

> **Project Track:** TA3. Chemotherapy Response and Tumor Suppression
> **Base Article:** Kisbenedek et al. (2026), "Physics-Informed Neural Network for Parameter Inference in a Tumor Model"
> **Team Members:** Sümeyra Şanlı, Zeynep Yıldırım, Korhan Kuş, Eren Gündoğmuş

1. Current Status & Initial Research

During the initial phase of our project, the team has focused on foundational literature review and theoretical understanding. Our progress to date includes:

- **Literature Review:** The foundational article by Kisbenedek et al. (2026) regarding the application of Physics-Informed Neural Networks (PINNs) in tumor models has been reviewed.
- **Theoretical Analysis:** The theoretical framework of quantifying chemotherapy response and tumor suppression using dynamic ODE models is currently being researched.
- **Brainstorming & Planning:** Initial brainstorming sessions were held to discuss potential novel contributions to the baseline methodology and to plan the project roadmap.

2. Short-Term Objectives

Prior to transitioning into the core implementation phase, our immediate objectives are:

- Clone the associated GitHub repository (`kisblilla/piecewise-pinn-tumor-growth`) and map the codebase architecture.
- Achieve full reproducibility of the baseline model within our local environments.
- Finalize the specific innovation/extension we will bring to the model and submit the Formal Proposal.